

UNIVERSITATEA TEHNICĂ A MOLDOVEI
FACULTATEA TELECOMUNICAȚII
SPECIALITATEA TEHNOLOGII ȘI SOFTWARE ÎN REȚELE DE
TELECOMUNICAȚII

PCLP

Lucru individual

A efectuat: Creciu Dumitrita , grupa TSRC-251

A verificat: Cerbu Olga, dr., conf. universitar

Chișinău 2026

Instalăm Elastic Search și Kibana cu ajutorul aplicației Docker

```
C:\Users\Emilia\Downloads\Telegram Desktop>docker compose up -d
time="2025-04-03T14:52:11+03:00" level=warning msg="C:\\Users\\Emilia\\Downloads\\Telegram Desktop\\docker-compose.yml:
the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 35/35
✔elasticsearch Pulled
✔kibana Pulled
✔grafana Pulled
```

☐ Only show running containers

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last sta	Actions
☐	kibana	f844dc88cc2f	kibana:9.3.0	5601:5601	N/A	2 days a	
☐	elasticsearch	c89e30bc79f1	elasticsearch	9200:9200	N/A	2 days a	

Observăm că totul s-a instalat corect

☐	Name	Tag	Image ID	Created	Size	Actions
☐	kibana	9.3.0	8f167d8b8c82	15 days ago	2.18 GB	
☐	elasticsearch	9.3.0	d6dbcf006047	15 days ago	2.14 GB	
☐	mcp/elasticsearch	latest	9c978b8277e3	8 months ago	32.4 MB	

Walkthroughs

Testăm în browser Elastic search pe *localhost:9200*

Observăm că totul merge bine

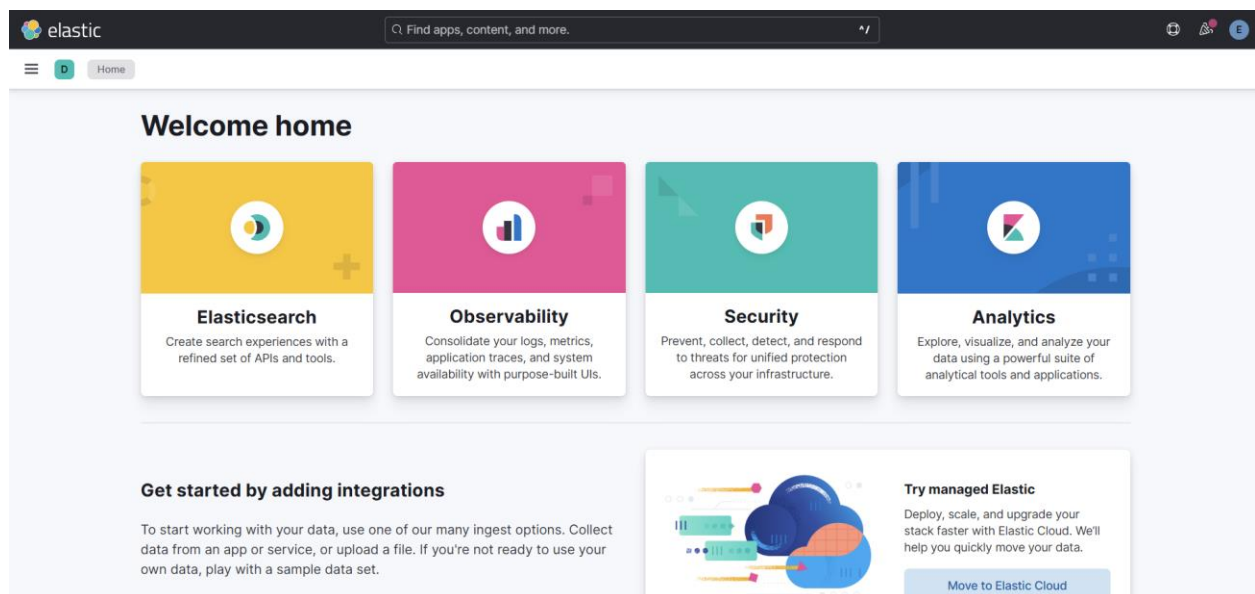
```
localhost:9200

Pretty-print ☐

{
  "name" : "DESKTOP-BMUVER3",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "RIjLlIbqTq6iM2fLZaZomQ",
  "version" : {
    "number" : "8.17.4",
    "build_flavor" : "default",
    "build_type" : "zip",
    "build_hash" : "c63c7f5f8ce7d2e4805b7b3d842e7e792d84dda1",
    "build_date" : "2025-03-20T15:39:59.811110136Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.0",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

Testăm în browser Kibana pe *localhost:5601*

Putem observa că am primit acces la Kibana



Pentru a crea diagramele am importat un fișier cu log-uri de pe github pentru sistemul de operare Linux

Name	Last commit message	Last commit date
..		
Linux_2k.log	Add log templates	2 years ago
Linux_2k.log_structured.csv	Add log templates	2 years ago
Linux_2k.log_templates.csv	Add log templates	2 years ago
README.md	Update dataset README	2 years ago

README.md

Am efectuat modificările pentru fișierul importat pentru că inițial erau datele implicite

×

Override settings

Number of lines to sample

2001

Data format

delimited

Delimiter

comma

Quote character

"

☒ Has header row

☐ Should trim fields

☒ Contains time field

Timestamp format

custom

×

Close

Apply

Am creat un nou index

Index Management

[Index Management docs](#)

[Indices](#) [Data Streams](#) [Index Templates](#) [Component Templates](#) [Enrich Policies](#)

Update your Elasticsearch indices individually or in bulk. [Learn more.](#)

☐ Include hidden indices ☐ Include rolup indices

Lifecycle status Lifecycle phase [Reload indices](#) [Create index](#)

☐ Name	Health	Status	Primaries	Replicas	Documents c...	Storage size	Data stream
-------------------------------	--------	--------	-----------	----------	----------------	--------------	-------------

No indices

You don't have any indices yet.

[Create index](#)

Console [Notebooks](#)

Am dat o denumire pentru index - test_logs și l-am importat

Linux_2k.log_structured.csv

Import data

[Simple](#) [Advanced](#)

Index name

test_logs

☒ Create data view

[Reset](#)

✓

✓

✓

✓

✓

File processed Index created Ingest pipeline created Data uploaded Data view created

Am testat câteva căutări cu ajutorul metodei Get

```
1 # Click the Variables button, above, to create your own variables.
2 GET /_search
3 {
4   "query": {
5     "match_all": {} // match_all
6   }
7 }
8 GET /test_logs/_search
9 {
10  "query": {
11    "bool": {
12      "must": [
13        { "match": { "Component": "sshd(pam_unix)" } },
14        { "match": { "PID": 19937 } },
15      ]
16      "range": {
17        "@timestamp": {
18          "gte": "2025-01-01T02:00:00.000+02:00",
19          "lte": "2025-01-01T15:30:00.000+02:00"
20        }
21      }
22    }
23  ]
24 }
25 }
26 }
```

Click to send request

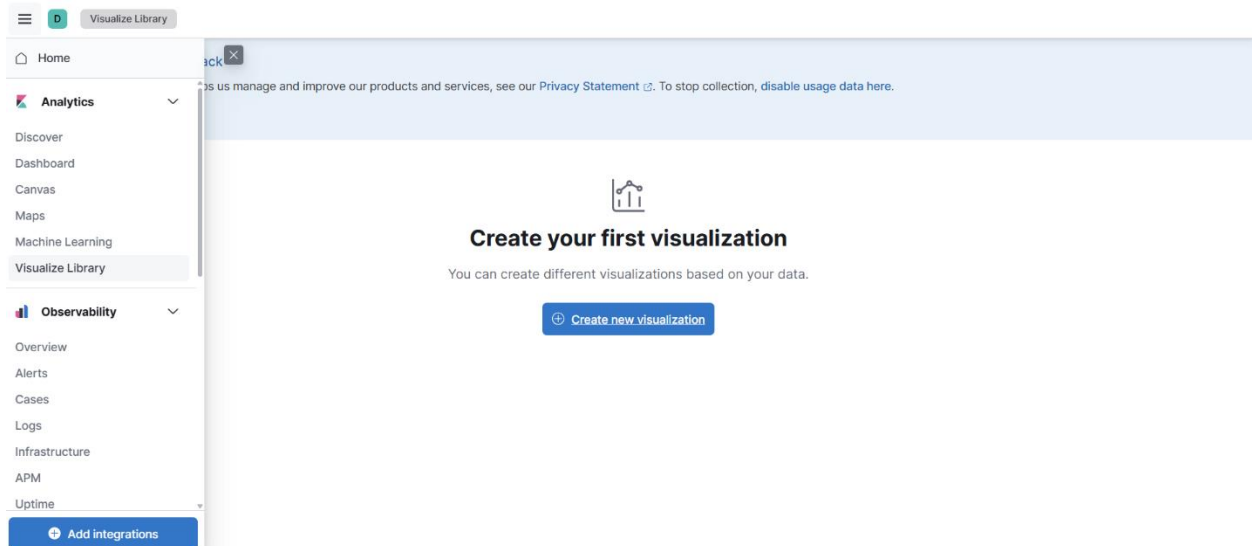
Observăm rezultatul după filtrele alese

```

35   {
36     "_index": "test_logs",
37     "_id": "LfiD-5UB508ESxVZ5T4N",
38     "_score": 3.0829926,
39     "source": {
40       "Month": "Jun",
41       "@timestamp": "2025-01-01T15:16:02.000+02:00",
42       "LineId": 3,
43       "Content": "authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser=
44         rhost=218.188.2.4",
45       "EventTemplate": "authentication failure; logname= uid=0 euid=0 tty=NODEVssh
46         ruser= rhost=<*>",
47       "EventId": "E16",
48       "Time": "15:16:02",
49       "Level": "combo",
50       "PID": 19937,
51       "Component": "sshd(pam_unix)",
52       "Date": 14
53     }
54   }
55 }

```

Am intrat la Visualize Library și am ales opțiunea Create new visualization



Am ales opțiunea Aggregation based

New visualization



Lens

Create visualizations with our drag and drop editor. Switch between visualization types at any time. *Recommended for most users.*



Maps

Create and style maps with multiple layers and indices.



TSVB

Perform advanced analysis of your time series data.



Custom visualization

Use Vega to create new types of visualizations. *Requires knowledge of Vega syntax.*



Aggregation based

Use our classic visualize library to create charts based on aggregations.

[Explore options →](#)

Tools



Text

Add text and images to your dashboard.



Input controls Deprecated

Input controls are deprecated and will be removed in a future version.

Want to learn more? [Read documentation](#)

Observăm că putem alege ce tip de diagrama dorim să creăm



Area

Emphasize the data between an axis and a line.



Data table

Display data in rows and columns.



Gauge

Show the status of a metric.



Goal

Track how a metric progresses to a goal.



Heat map

Display values as colors in a matrix.



Horizontal bar

Present data in horizontal bars on an axis.



Line

Display data as a series of points.



Metric

Show a calculation as a single number.



Pie

Compare data in proportion to a whole.



Tag cloud

Display word frequency with font size.



Timelion

Show time series data on a graph.



Vertical bar

Present data in vertical bars on an axis.

Am ales prima data diagrama de tip line

Alegem test_logs care conține datele noastre si configuram

New Line / Choose a source

[← Select a different visualization](#)

Sort ▾
Types 2 ▾


test_logs

Alegem caracteristicile dorite

test_logs

Data Metrics & axes Panel settings

Metrics

> [Y-axis Count](#)

+ Add

Buckets

∨ X-axis

👁 = ✕

Aggregation

[Date Histogram help](#)

Date Histogram

Field

@timestamp

Minimum interval

Second

Select an option or create a custom value. Examples: 30s, 20m, 24h, 2d, 1w, 1M

Sub aggregation

[Terms help](#)

Terms

Field

PID

Order by

Metric: Count

Order

Size

Descending

5

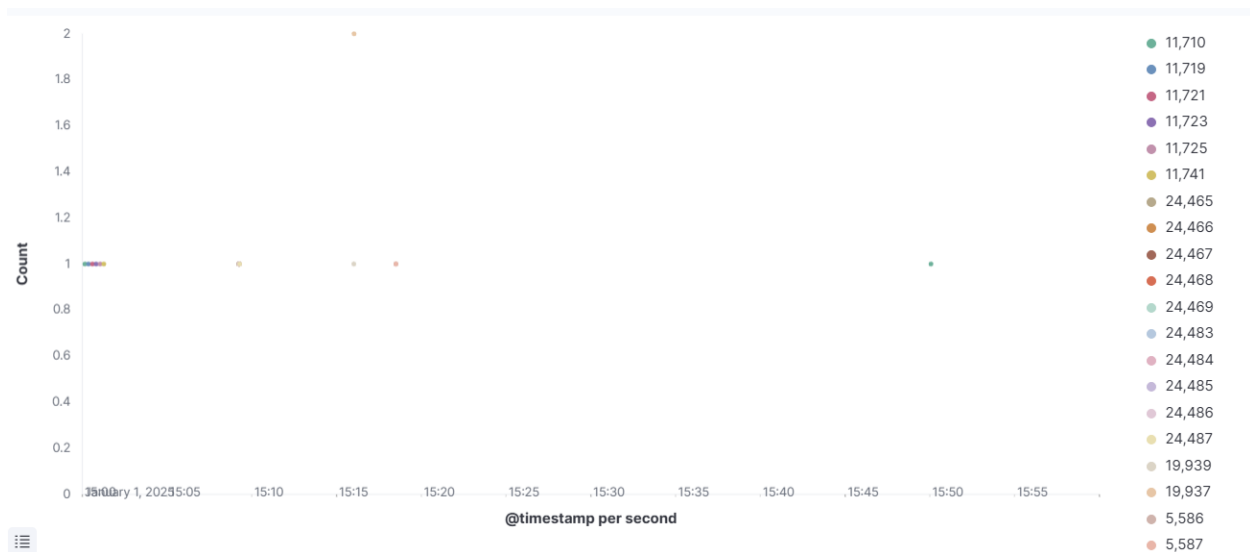
Alegem data pentru care se vor efectua căutările



Jan 1, 2025 @ 15:00:00.000 → Jan 1, 2025 @ 16:00:00.000

🔄 Refresh

Rezultatul



Am ales a doua oară diagrama de tip pie-chart și am ales configurările dorite

Jan 1, 2025 @ 15:00:00.000 → Jan 1, 2025 @ 16:00:00.000

Metrics

▼ Slice size

Aggregation

Sum help [↗](#)

Sum



Field

PID



Custom label

Split chart

Rows

Columns

Aggregation

Date Histogram help

Date Histogram

Field

@timestamp

Minimum interval

Second

Select an option or create a custom value. Examples: 30s, 20m, 24h, 2d, 1w, 1M

Rezultatul reprezintă mai multe diagrame, fiecare pentru un anumit timp din intervalul selectat anterior



Rezultatul final reprezintă cele două diagrame

